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Organic food consumption and the incidence of type 2 diabetes mellitus in the Danish Diet, Cancer and Health cohort

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ABSTRACT

Aim: To investigate the association between organic food consumption and the incidence of type 2 diabetes mellitus.

Methods: Among 41,286 cohort participants, aged 50–65 years, organic food consumption of vegetables, fruits, dairy products, eggs, meat, and cereal products, was summarized into an organic food score evaluated as never, low, medium and high consumption and as continuous intake. During follow-up, 4,843 cases were identified in the National Diabetes Register. Organic food consumption was associated to the disease incidence in Cox regression models.

Results: Organic food consumption was linearly associated with a lower incidence of type 2 diabetes mellitus (Women, HR: 0.94, 95% CI: 0.89–1.00, Men, HR: 0.95, 95% CI: 0.90–1.00). Organic food consumption frequency, compared to never consumption, showed HRs below 1.00 for both women (medium intake HR: 0.96, 95% CI: 0.84–1.10, high intake HR: 0.88, 95% CI: 0.74–1.05) and men (low intake, HR: 0.95, 95% CI: 0.85–1.05, medium intake, HR: 0.92, 95% CI: 0.83–1.03, high intake, HR: 0.89, 95% CI: 0.75–1.05) but were not statistically significant. Similar patterns were observed with consumption of the specific organic food groups for women, but not for men.

Conclusions: Organic food consumption was associated with a suggested lower incidence of type 2 diabetes mellitus.

1. Introduction

The global prevalence of diabetes was 537 million in 2021 and is predicted to further increase during the coming years [1]. About 90% of all diabetes cases are type 2 diabetes mellitus, which is a complex disease influenced by genetics, epigenetics and lifestyle factors [2]. Metabolic risk factors as well as unhealthy diet and lifestyle are established modifiable risk factors of type 2 diabetes mellitus [2,3]. A dietary factor that has attracted less attention regarding incidence of type 2 diabetes mellitus is consumption of organic foods.

Expected positive health effects is a major reason when consumers purchase organic foods [4]. Suggested explanations for a potentially beneficial health effect of consuming organically produced foods compared to conventionally produced foods relate to lower levels of pesticide residues [5], antibiotic resistant bacteria, and potentially higher levels of phytochemicals and a more favorable fatty acid composition [6,7]. Both animal and epidemiological studies have shown that exposure to pesticides have the potential to increase diabetes risk by altering glucose and lipid metabolism, and induce insulin resistance [8,9]. Polyphenols, and especially flavonoids have been associated with lower risk of type 2 diabetes mellitus [10] which might be explained by flavonoids anti-oxidative properties as well as potential influence on insulin receptor activation that improves insulin sensitivity and glucose uptake [10,11].

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Only one previous prospective study has examined the association between the proportion of organic food in the diet and the incidence of type 2 diabetes mellitus. The study by Kesse-Guyot found a 35% lower rate of type 2 diabetes mellitus among participants with the highest proportion of organic food in the diet compared to those with the lowest proportion [12]. In a cross sectional study examining associations between organic food purchase and risk of type 2 diabetes mellitus, lower odds of type 2 diabetes mellitus was observed among persons who said yes to any purchase of organic foods in the past 30 days compared to no purchase [13].

In the present study, we evaluated if organic food consumption, overall and by specific food groups, was associated with the incidence of diabetes in women and men.

2. Methods

2.1. Setting

The Danish Diet, Cancer and Health cohort includes 57,053 Danish men and women enrolled between December 1993 and May 1997. Participants were invited to participate in the study if they were in the age group 50–65 years, lived in the areas of Copenhagen or Aarhus and had no previous cancer diagnosis registered in the Danish Cancer Registry at the time of invitation [14]. The study was conducted according to the guidelines in the Declaration of Helsinki [15] and all procedures involving study participants were approved by regional ethical committees on human studies in Aarhus and Copenhagen and by the Danish Data Protection Agency.

2.2. Data collection

Between December 1993 and May 1997, participants visited a study center in one of the two largest cities in Denmark, i.e. in either Aarhus or Copenhagen where anthropometric measurements and biological samples were collected. Moreover, participants filled in a 192-item food frequency questionnaire (FFQ) and a lifestyle questionnaire. Between 1999 and 2002, corresponding to five years after baseline, participants received a follow-up lifestyle questionnaire and an updated FFQ with additional questions about organic food consumption.

2.3. Exposure

Information about organic food consumption was obtained from the follow-up questionnaires completed between 1999 and 2002. Participants reported organic food consumption of vegetables, fruits, dairy products, eggs, meat, and bread and cereal products. Organic food consumption was reported in frequencies specified as never, sometimes, often and always. The reporting of consumption of the six organic food groups was converted into a total organic food score, by adding up the values corresponding to never=1, sometimes=2, often=3, and always=4, thus ranging from 6 to 24. The total organic food score was then categorized into never consumption of organic foods (organic food score of 6), low organic food consumption (organic food score of 7–12), medium organic food consumption (organic food score of 13–18) and high organic food consumption (organic food score 19-24).

2.4. Covariates

We used information about educational level (short <8 years, medium 8–10 years, long >10 years) retrieved from the baseline questionnaire. From the follow-up questionnaires, we used information about body mass index (BMI), physical activity, smoking (never/former/current), alcohol- and dietary intake. From the lifestyle questionnaire, participants were asked to report the duration and frequency of light, moderate and vigorous physical activity ranging from one or less than one per month to 11 or more times per week for the summer and winter half-year. If the average sum of moderate, and vigorous physical activity for the summer and winter half year were equal or greater to half an hour per week, participants were considered to participate in sports. We also included information about duration of physical activity among the actives (hours/week). From the FFQ, participants were asked to report their average intake of different foods and beverages over the past year in frequencies ranging from less than once per month and up to eight or more times per day. The daily intake of foods, nutrients, and alcohol was calculated using the program FoodCalc [16].

2.5. Identification of cases

By record linkage using the Danish civil registration number, we identified diabetes cases diagnosed from 1990 to December 2012 from the Danish National Diabetes Register. Participants were registered with a diabetes diagnosis in the National Diabetes Register if one of the six following criteria was fulfilled: 1) Registered with diabetes with one of the codes; ICD-10: DE10-14, DH36.0, DO24 (excluding DO24.4) and ICD-8 (prior to 1994): 249, 250, 2) Registration of chiropody as a diabetic patient in the National Health Service Register (NHSR), 3) Registered with five blood glucose measurements within a one year period in NHSR, 4) Registered with two blood glucose measurements per year for a period of five years in NHSR, 5) Registered purchase of oral antidiabetic drugs (OAD) in the Danish National Prescription Registry (DNPR), 6) Endorsement of prescribed insulin recorded in DNPR (17). The National Diabetes Register does not distinguish between type 1 diabetes and type 2 diabetes mellitus. As the cohort participants were at minimum 53 years at entrance into the current study we assume that the majority of cases were type 2 diabetes mellitus and refer to that hereafter [18].

2.6. Study population

The current study is based on the 44,872 participants who responded to the follow-questionnaires including information about organic food consumption. Of these, 1,965 were excluded due to a diabetes diagnosis registered in the Danish Diabetes Register before completion of the follow-up questionnaires. Further, 1,621 participants were excluded due to missing information about organic food consumption (n=1,448) or different covariates (n=173) leaving 41,286 for the present study.

2.7. Statistical analyses

Descriptive statistics of lifestyle and sociodemographic characteristics are presented separately for women and men and by the level of organic food consumption. Cox proportional hazards models were used to estimate hazard ratios (HR) and 95% confidence intervals (CI) for the associations between organic food consumption, overall and by specific food groups, and the incidence of type 2 diabetes mellitus. The analyses were corrected for delayed entry so individuals were only considered at risk from the age at completion of follow-up questionnaires until age at diabetes diagnosis, death, emigration or end of follow-up (December 31, 2012), whichever came first. Organic food consumption was evaluated as an overall organic food score in frequencies specified as never, low, medium and high consumption, and by organic consumption frequency of vegetables, fruits, dairy products, eggs, meat, and bread and cereal products. Never consumers of organic foods overall and never consumers of the specific organic foods were used as the reference categories. Moreover, the organic food score ranging from 6 to 24 was evaluated as a continuous exposure with estimates per 6-point increments provided, and organic food consumption of the specific food groups as ordinal variables ranging from 1 to 4. We evaluated two models with covariates selected on a priori approach based on existing literature [2,3] and analyzed associations separately for women and men. A crude model (Model 1), adjusted for age as the underlying time

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Table 1
Characteristics of the cohort stratified by sex and organic food score consumption.

	Women					Men				
	No. (%) or median (IQR)					No. (%) or median (IQR)				
Organic food score consumption	Total cohort	Never	Low	Medium	High	Total cohort	Never	Low	Medium	High
No. of participants	22,082 (100%)	2,676 (12%)	8,040 (36%)	8,798 (40%)	2,568 (12%)	19,204 (100%)	3,295 (17%)	7,876 (41%)	6,473 (34%)	1,567 (8%)
Age at study entry, years	61 (58–66)	62 (58–66)	61 (58–66)	61 (58–65)	61 (58–65)	61 (58–65)	62 (58–66)	61 (58–65)	61 (58–65)	61 (58–65)
Diabetes, no.	2,222	310 (12%)	858 (11%)	836 (10%)	218 (8%)	2,622	522 (16%)	1,099 (14%)	816 (13%)	184 (12%)
Age at diabetes, years	68 (64–72)	68 (64–72)	68 (64–72)	68 (64–72)	68 (65–72)	68 (64–72)	68 (64–72)	68 (64–71)	68 (64–71)	68 (64–71)
BMI, kg/m2	25 (22–27)	25 (23–28)	25 (22–28)	24 (22–27)	24 (22–27)	26 (24–28)	26 (24–29)	26 (24–28)	26 (24–28)	26 (24–28)
Physical activity										
– Yes	10,816 (49%)	1,098 (41%)	3,802 (47%)	4,541 (52%)	1,375 (54%)	7,973 (41%)	1,115 (34%)	3,209 (41%)	2,936 (45%)	713 (46%)
– No	11,266 (51%)	1,578 (59%)	4,238 (53%)	4,257 (48%)	1,193 (46%)	11,231 (59%)	2,180 (66%)	4,667 (59%)	3,537 (55%)	847 (54%)
Duration among actives, hours/week	1 (1–3)	1 (1–2)	1 (1–2)	1 (1–3)	2 (1–3)	2 (1–3)	2 (1–4)	2 (1–3)	2 (1–3)	2 (1–4)
Alcohol, g/day	10 (4–19)	9 (3–18)	10 (4–20)	11 (4–20)	11 (4–19)	20 (10–40)	19 (9–40)	20 (10–41)	21 (11–40)	20 (11–40)
Educational length										
– Short	6,342 (29%)	1,144 (43%)	2,512 (31%)	2,131 (24%)	537 (21%)	6,232 (33%)	1,369 (42%)	2,631 (33%)	1,801 (28%)	409 (26%)
– Medium	11,404 (52%)	1,236 (46%)	4,280 (53%)	4,569 (52%)	1,299 (51%)	8,152 (42%)	1,317 (40%)	3,421 (44%)	2,784 (43%)	619 (40%)
– Long	4,387 (20%)	296 (11%)	1,248 (16%)	2,098 (24%)	732 (28%)	4,863 (25%)	609 (18%)	1,824 (23%)	1,888 (29%)	532 (34%)
Smoking										
– Never	10,062 (45%)	1,144 (43%)	3,747 (47%)	4,018 (46%)	1,120 (44%)	5,252 (27%)	846 (26%)	2,190 (28%)	1,768 (27%)	441 (28%)
– Former	6,499 (28%)	677 (25%)	2,206 (27%)	2,728 (31%)	878 (34%)	8,084 (42%)	1,268 (38%)	3,255 (41%)	2,843 (44%)	699 (45%)
– Current	5,572 (25%)	845 (32%)	2,087 (26%)	2,052 (23%)	570 (22%)	5,911 (31%)	1,181 (36%)	2,431 (31%)	1,862 (29%)	420 (27%)
Diet, g/day										
Vegetables	156 (103–226)	122 (79–181)	141 (93–202)	171 (116–239)	195 (137–276)	140 (93–204)	113 (71–167)	133 (88–193)	158 (108–221)	178 (123–254)
Fruits	279 (136–408)	217 (102–334)	263 (126–370)	294 (157–440)	309 (185–473)	188 (94–325)	137 (62–294)	170 (87–318)	220 (110–343)	225 (128–374)
Dairy products	312 (172–572)	288 (127–555)	295 (154–544)	330 (195–589)	346 (207–612)	307 (163–582)	281 (133–570)	300 (153–573)	323 (192–590)	341 (211–606)
Eggs	18 (12–27)	18 (11–28)	18 (11–27)	18 (12–27)	18 (12–27)	21 (14–31)	21 (14–32)	21 (14–30)	21 (14–31)	22 (15–32)
Meat	111 (85–143)	114 (88–146)	113 (88–144)	111 (85–144)	101 (75–133)	168 (131–214)	175 (135–221)	170 (134–216)	165 (130–210)	155 (117–204)
Bread and cereal products	166 (126–212)	149 (110–192)	162 (123–207)	171 (132–219)	176 (133–223)	204 (155–259)	190 (138–242)	201 (153–255)	211 (162–266)	218 (169–271)

Abbreviations: No., Number, IQR, Inter quartile range, BMI, Body mass index.

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scale, and time since completion of the follow-up questionnaires to secure comparison of two persons with the same duration of exposure. The primary Model 2 further adjusted for educational length (short <8 years, medium 8–10 years, long 10> years), BMI (kg/m2), physical activity (yes/no and duration among actives, hours/week), smoking habits (never/former/current), and intake of alcohol (g/day). In the analyses with organic food consumption of the specific food groups, we adjusted for the intake of the specific food group that the organic exposure group was covering. E.g. in the analyses evaluating the association between organic food consumption of fruits and type 2 diabetes mellitus, we adjusted for the intake of fruits (g/day). We also evaluated whether adjustment for dietary intake (g/day) of major food groups including vegetables, fruits, dairy products, red meat, processed meat, poultry, whole grain products, refined grain products, nuts and seeds, soft drinks, and confectionery yielded different results and whether associations differed by overall diet quality. To evaluate diet quality, we constructed a diet score including the following cut points based on Danish dietary guidelines; <500 g meat per week, >200 g fish per week, >75 g whole grain per day, and >600 g fruit and vegetables per day. Persons were scored 0 if they did not fulfill a specific recommendation and 1 if they fulfilled the recommendation, thus the diet score ranged from 0 to 4. We then divided persons in three strata based on the following diet-score categories: 0–1, 2, 3–4. The analyses with adjustment for dietary intake and modification by diet score did not yield different results and are therefore not included in the manuscript. In addition, for overall organic food consumption frequency, we estimated the absolute risk of type 2 diabetes mellitus non-parametrically using the Aalen-Johansen estimator [19]. Furthermore, we calculated the average age-specific standardized absolute risk estimates based on Cox regression models [20,21] including the same covariates as for model 2 both for the outcome of interest (type 2 diabetes mellitus) and the competing event (death). In these models BMI and alcohol intake was categorized in four groups as (<18.5, 18.5–24.9, 25–29.9, ≥ 30 kg/m2) and (<10, 10–19.9, 20–29.9, ≥ 30 g/day) respectively and physical activity represented by the indicator variable (yes/no). We used age as the underlying time-scale with delayed entry at 60 years to avoid high variability in younger ages due to the small number at risk before age 60. Finally, we conducted a sensitivity analysis and evaluated the association between the individual organic food groups of the organic food score as continuous variables and incidence of type 2 diabetes mellitus by mutually adjusting for the other organic food groups with the same adjustment strategy as for the primary Model 2 (Supplementary Table S1). Continuous covariates were tested for linearity by Wald tests using linear splines with boundaries at 25th, 50th and 75th percentiles among cases and for BMI, we used cut points at 18.5, 24.9 and 29.9 kg/m^2. P-values below 0.05 were interpreted as deviation from linearity. BMI and alcohol were included in the model as linear splines. For alcohol, we placed the knots per 12-gram alcohol corresponding to one unit. Linearity of cereals and fruit intake deviated in the analyses for women, and were included in these specific models as linear splines with knots placed at the quartile intake among cases. Similarly, for men the intake of vegetables deviated from linearity in the analyses, and were included in the model as a linear spline with knots placed at the quartile intake among cases. The proportional hazard assumption was tested by Schoenfeld residuals [22] and no visual deviation from proportionality over time was observed.

3. Results

3.1. Characteristics of the study population

During the study period, 4,843 incident type 2 diabetes mellitus cases were registered (2,222 women and 2,621 men) among the 41,286 cohort participants. Table 1 shows characteristics of the cohort stratified by sex and across categories of overall organic food consumption. The majority of women (40%) had a medium level of organic food consumption whereas the majority of men (41%) had a low level of organic food consumption. Women in the highest organic food consumption category had a slightly lower BMI, were more likely to participate in sports, had higher educational level, and were less likely to smoke compared to women in the lower organic food consumption categories. Similar patterns were observed for men, however, no differences were observed for BMI across categories of organic food consumption (Table 1). Women in the highest organic food consumption category compared to never consumers had higher intakes of vegetables, fruits, dairy products, poultry, fish, bread and cereal products with the major intake consisting of whole grain products, nuts and seeds, and alcohol, and had lower intakes of red and processed meat and refined grain products. No markedly differences were observed for intakes of eggs, soft drinks, and confectionery (Fig. 1). Similar patterns were observed for men, however men in the highest organic food score consumption category compared to never consumers had lower intake of soft drinks and confectionery but no differences were observed for intake of alcohol (Fig. 1).

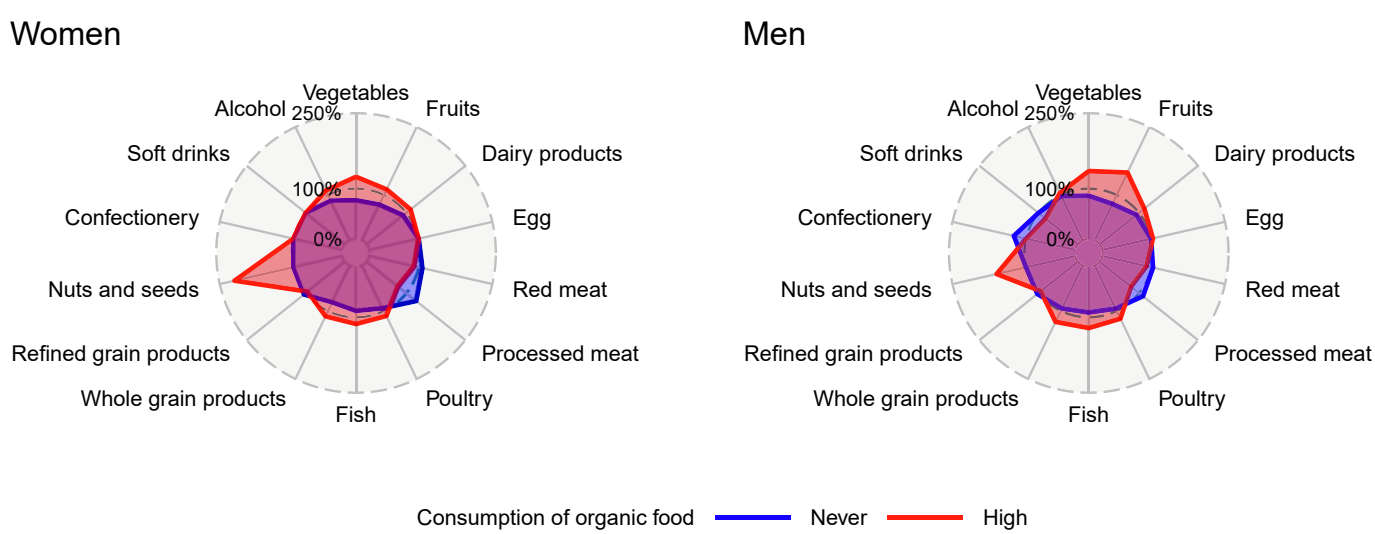

Fig. 1. Radar chart showing percentage difference in intakes of different food groups among never and high consumers of organic foods separately for women and men. 100% is the median intake of women (n = 22,082) and men (n = 19,204).

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Table 2
Hazard ratios and 95% confidence intervals of incidence of type 2 diabetes mellitus by organic food consumption, overall, and by specific food groups for women.

Women	n = 22,082			
	Model 1 (crude)		Model 2	
	HR	95% CI	HR	95% CI
Organic food score consumption				
Never 6	1.00		1.00	
Low 7–12	0.92	0.81–1.05	1.03	0.90–1.17
Medium 13–18	0.81	0.71–0.93	0.96	0.84–1.10
High 19-24	0.72	0.61–0.86	0.88	0.74–1.05
Continuous with a 6-point increment	0.87	0.83–0.93	0.94	0.89–1.00
Organic vegetables				
Never	1.00		1.00	
Sometimes	0.89	0.80–0.99	0.98	0.89–1.09
Often	0.79	0.70–0.89	0.93	0.82–1.05
Always	0.78	0.63–0.96	0.90	0.72–1.11
Ordinal variable	0.90	0.86–0.95	0.96	0.91–1.01
Organic fruits				
Never	1.00		1.00	
Sometimes	0.90	0.82–0.99	1.00	0.91–1.10
Often	0.84	0.75–0.95	0.97	0.86–1.10
Always	0.80	0.62–1.00	0.83	0.65–1.05
Ordinal variable	0.92	0.87–0-97	0.97	0.92–1.02
Organic dairy products				
Never	1.00		1.00	
Sometimes	1.00	0.90–1.11	1.05	0.95–1.17
Often	0.82	0.73–0.92	0.94	0.83–1.06
Always	0.83	0.72–0.96	0.90	0.77–1.04
Ordinal variable	0.92	0.88–0.96	0.96	0.92–1.00
Organic eggs				
Never	1.00		1.00	
Sometimes	1.00	0.90–1.12	1.05	0.94–1.18
Often	0.85	0.75–0.97	0.94	0.83–1.07
Always	0.74	0.66–0.84	0.86	0.76–0.97
Ordinal variable	0.90	0.90–0.93	0.94	0.91–0.98
Organic meat				
Never	1.00		1.00	
Sometimes	0.87	0.79–0.95	0.92	0.84–1.01
Often	0.83	0.73–0.94	0.88	0.77–1.00
Always	1.10	0.84–1.44	1.16	0.89–1.51
Ordinal variable	0.93	0.88–0.99	0.96	0.91–1.02
Organic bread and cereals				
Never	1.00		1.00	
Sometimes	0.94	0.85–1.04	1.00	0.90–1.10
Often	0.82	0.73–0.93	0.96	0.85–1.08
Always	0.84	0.72–0.98	0.98	0.83–1.14
Ordinal variable	0.93	0.89–0.97	0.99	0.94–1.03

Model 1: crude model adjusted for age as the underlying time scale and age at entry into the study.
Model 2: Model 1 further adjusted for smoking (never/former/current), alcohol consumption (g/day), educational level (short, medium, long), BMI (kg/m2) and physical activity (yes/no, and duration among actives hours/week). For the analyses with organic food consumption of the specific food groups, the model is further adjusted for the dietary intake (g/day) that the organic exposure group covers. E.g. in the model with organic vegetables as the exposure, adjustment for the intake of vegetables has been made.

3.2. Organic food consumption and the incidence of type 2 diabetes mellitus

Table 2 shows the crude and adjusted associations between organic food consumption overall and by specific food groups and the incidence of type 2 diabetes mellitus for women. Compared to never consumption of organic foods overall, medium intake (HR: 0.96, 95% CI: 0.84–1.10), and high intake (HR: 0.88, 95% CI: 0.74–1.05) of organic foods showed HRs below 1.00 but were not statistically significant (Model 2). Continuous organic food score consumption was associated with an estimated 6% lower incidence of type 2 diabetes mellitus per 6-point increments in score (HR: 0.94, 95% CI: 0.89–1.00) (Table 2). Similar patterns were observed with consumption of the specific organic food groups, except for meat where always consumption of organic meat compared to never consumption was associated with an insignificant higher incidence of diabetes.

Table 3 shows the crude and adjusted associations between organic food consumption overall and by specific food groups and the incidence of type 2 diabetes mellitus for men. Compared to never consumption of organic foods overall, low intake (HR: 0.95, 95% CI: 0.85–1.05), medium intake (HR: 0.92, 95% CI: 0.83–1.03), and high intake (HR: 0.88, 95% CI: 0.75–1.05) of organic foods showed HRs below 1.00 but were not statistically significant (Model 2). Continuous organic food score consumption was associated with a 5% lower incidence of type 2 diabetes mellitus per 6-point increments in score (HR: 0.95, 95% CI: 0.90–1.00) (Table 3). No clear pattern was observed when evaluating organic food consumption of the specific food groups and incidence of type 2 diabetes mellitus (Table 3).

Supplementary Table S1 shows the association between the individual organic food groups of the organic food score as continuous variables and incidence of type 2 diabetes mellitus by mutually adjusting for the other organic food groups based on the primary model (model 2). Similar to the frequency consumption of organic eggs compared to never consumption of organic eggs (Table 2 and Table 3), inverse associations were observed between the intake of organic eggs per one unit increase and the incidence of type 2 diabetes mellitus (women, HR: 0.94, 95% CI:

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Table 3
Hazard ratios and 95% confidence intervals of incidence of type 2 diabetes mellitus by organic food consumption, overall, and by specific food groups for men.

Men	n = 19,204			
	Model 1 (crude)		Model 2	
	HR	95% CI	HR	95% CI
Organic food score consumption				
Never 6	1.00		1.00	
Low 7–12	0.86	0.78–0.96	0.95	0.85–1.05
Medium 13–18	0.78	0.70–0.87	0.92	0.83–1.03
High 19-24	0.72	0.61–0.85	0.89	0.75–1.05
Continuous with a 6-point increment	0.88	0.83–0.92	0.95	0.90–1.00
Organic vegetables				
Never	1.00		1.00	
Sometimes	0.91	0.83–1.00	1.02	0.93–1.12
Often	0.81	0.73–0.91	0.98	0.87–1.09
Always	0.80	0.64–0.99	0.99	0.80–1.23
Ordinal variable	0.91	0.87–0.95	0.99	0.94–1.04
Organic fruits				
Never	1.00		1.00	
Sometimes	0.86	0.79–0.94	0.93	0.85–1.01
Often	0.86	0.77–0.97	0.96	0.86–1.08
Always	0.91	0.71–1.16	0.99	0.77–1.27
Ordinal variable	0.93	0.88–0.98	0.98	0.93–1.03
Organic dairy products				
Never	1.00		1.00	
Sometimes	0.92	0.84–1.01	0.97	0.88–1.06
Often	0.81	0.73–0.90	0.94	0.84–1.05
Always	0.75	0.65–0.86	0.88	0.76–1.01
Ordinal variable	0.90	0.87–0.94	0.96	0.92–1.00
Organic eggs				
Never	1.00		1.00	
Sometimes	0.89	0.81–0.97	0.93	0.84–1.02
Often	0.85	0.76–0.95	0.94	0.84–1.05
Always	0.70	0.62–0.79	0.81	0.72–0.91
Ordinal variable	0.90	0.87–0.93	0.94	0.91–0.98
Organic meat				
Never	1.00		1.00	
Sometimes	0.93	0.86–1.01	1.00	0.92–1.09
Often	0.92	0.81–1.04	0.99	0.87–1.13
Always	0.98	0.74–1.30	1.01	0.76–1.33
Ordinal variable	0.96	0.91–1.01	1.00	0.95–1.05
Organic bread and cereals				
Never	1.00		1.00	
Sometimes	0.95	0.87–1.03	1.02	0.94–1.12
Often	0.79	0.71–0.89	0.94	0.84–1.05
Always	0.79	0.67–0.93	0.95	0.80–1.12
Ordinal variable	0.91	0.87–0.95	0.98	0.94–1.02

Model 1: crude model adjusted for age as the underlying time scale and age at entry into the study.
Model 2: Model 1 further adjusted for smoking (never/former/current), alcohol consumption (g/day), educational level (short, medium, long), BMI (kg/m2) and physical activity (yes/no, and duration among actives hours/week). For the analyses with organic food consumption of the specific food groups, the model is further adjusted for the dietary intake (g/day) that the organic exposure group covers. E.g. in the model with organic vegetables as the exposure, adjustment for the intake of vegetables has been made.

0.89–0.99, men, HR: 0.93, 95% CI: 0.88–0.98). No association was observed between the intake of organic vegetables, fruits, dairy products, meat, and bread and cereal products and the incidence of type 2 diabetes mellitus (Supplementary Table S1).

3.3. Absolute risk

Fig. 2 shows the crude and standardized estimates of the absolute risk of type 2 diabetes mellitus across organic food score consumption categories according to age. Compared to never consumption of organic foods, high consumption of organic foods at age 70 was associated with a 4% lower risk of type 2 diabetes mellitus for both women (high: 11.3% versus never: 7.2%) and men (high: 14.3% versus never: 10.0%) in the crude model. The standardized model considering covariates showed a 1% and 1.6% difference in risk of type 2 diabetes mellitus at age 70 for women (high: 9.0% versus never: 7.9%) and men (high: 12.7% versus 11.1%) respectively (Fig. 2).

4. Discussion

In a large cohort of Danish women and men, overall organic food consumption was associated with a suggested lower incidence of type 2 diabetes mellitus. Both women and men with high compared to never consumption of organic eggs had a lower incidence of type 2 diabetes mellitus, after adjustment for various potential confounders.

A limitation was exposure assessment through FFQ. Participants were asked to report their intake of organic foods in frequencies specified as never, sometimes, often and always which are subjective measures and does not provide an estimate of the exact quantity of organic food consumed. Moreover, the supply of organic foods in Denmark have increased over the last decades [23], implying that the proportion of persons consuming organic foods may have increased, and that persons who consumed organic foods at study entry may have been extra conscious about their food choices, as well as previous non-consumers of organic food afterwards may have changed. It has previously been observed, in both our cohort and in other studies, that persons

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consuming organic foods have healthier diet and lifestyle patterns [24–26] and that these exposures are associated with a lower risk of type 2 diabetes mellitus [2,3]. This was supported in our study as adjustment for potential confounders including sociodemographic and lifestyle factors attenuated the associations observed between organic food consumption and lower incidence of type 2 diabetes mellitus (Table 2 and Table 3), however dietary intake of major food groups did not affect the associations markedly (data not shown). Information about diabetes diagnosis was retrieved from the Danish National Diabetes Register [17] which limit the risk of misclassification of cases compared to self-reporting, though the National Diabetes Register does not disentangle between diabetes types. However, diabetes data from the Register of Chronic Diseases [27], which disentangle between type 1 and type 2 diabetes mellitus, revealed almost the same number of cases for the present study period, whereof less than 2% of cases were diagnosed with type 1 diabetes (data not shown). Despite the above limitations, the current study also have several strengths such as the large number of cases, long follow-up, and few persons lost to follow-up, which provide sufficient power for the analyses.

Both of the previous studies by Kesse-Guyot [12] and Sun et al. [13] found a statistical significant association between organic food consumption and lower risk of type 2 diabetes mellitus, whereas we only observed a suggested lower incidence of type 2 diabetes mellitus with frequency consumption and continuous intake of organic foods. Organic foods contain lower levels of pesticide residues compared to conventional produced foods [5]. It has been demonstrated in animal studies that organophosphate pesticides can induce hyperglycemia by metabolic pathways in liver, skeletal muscles and brain [28] and insulin resistance due to disturbances in lipid metabolism, activation of inflammatory cytokines and oxidative stress [9]. Few epidemiological studies have investigated exposure to organophosphate pesticides, but two studies found that organophosphate pesticide application was associated with increased risk of diabetes [29,30]. A recently published study by Rebouillat et al. found that dietary exposure to pesticides was associated with risk of type 2 diabetes mellitus [31]. In addition, some studies have observed higher levels of phytochemicals in organically produced plants foods [32] and flavonoids have been associated with lower incidence of diabetes in epidemiological studies [10,33]. A potential lower exposure to pesticide residues and higher exposure to flavonoids may be a biological plausible explanation for the suggested lower incidence of type 2 diabetes mellitus.

We did also investigate if a potential association between organic food consumption and diabetes risk was driven by organic food consumption of specific food groups, to better understand the potential underlying mechanism. For these analyses, we further adjusted for the actual intake of the specific food group, as we wanted to compare persons with the same intake of the food group and rule out a potential effect of a higher or lower intake of that specific food group on type 2 diabetes mellitus risk. We observed that organic food consumption frequency of eggs but not vegetables, fruits, dairy products, meat, and bread and cereal products, was associated with a lower incidence of type 2 diabetes mellitus. The association with organic food consumption of the specific food groups must be interpreted with caution, as we have previously observed that the intake of the different organic food groups was highly correlated (data not shown). Therefore, the observed association with organic food consumption of eggs in the current study could potentially be due to a generally high intake of the other organic food groups. To evaluate the independent association of each organic food group included in the organic food score, we made a sensitivity analysis and mutually adjusted for intake of the other organic food groups and actual intake of the corresponding food group for the primary model 2. After mutual adjustment, the associations with organic food consumption of the specific food groups attenuated and pointed towards no association, which support the correlation between the organic food groups (Supplementary Table S1). However, the association between organic food consumption of eggs and incidence of type 2 diabetes mellitus remained negative (women, HR: 0.94, 95% CI: 0.89–0.99, men, HR: 0.93, 95% CI: 0.88–0.98). The cross-sectional study by Sun et al.

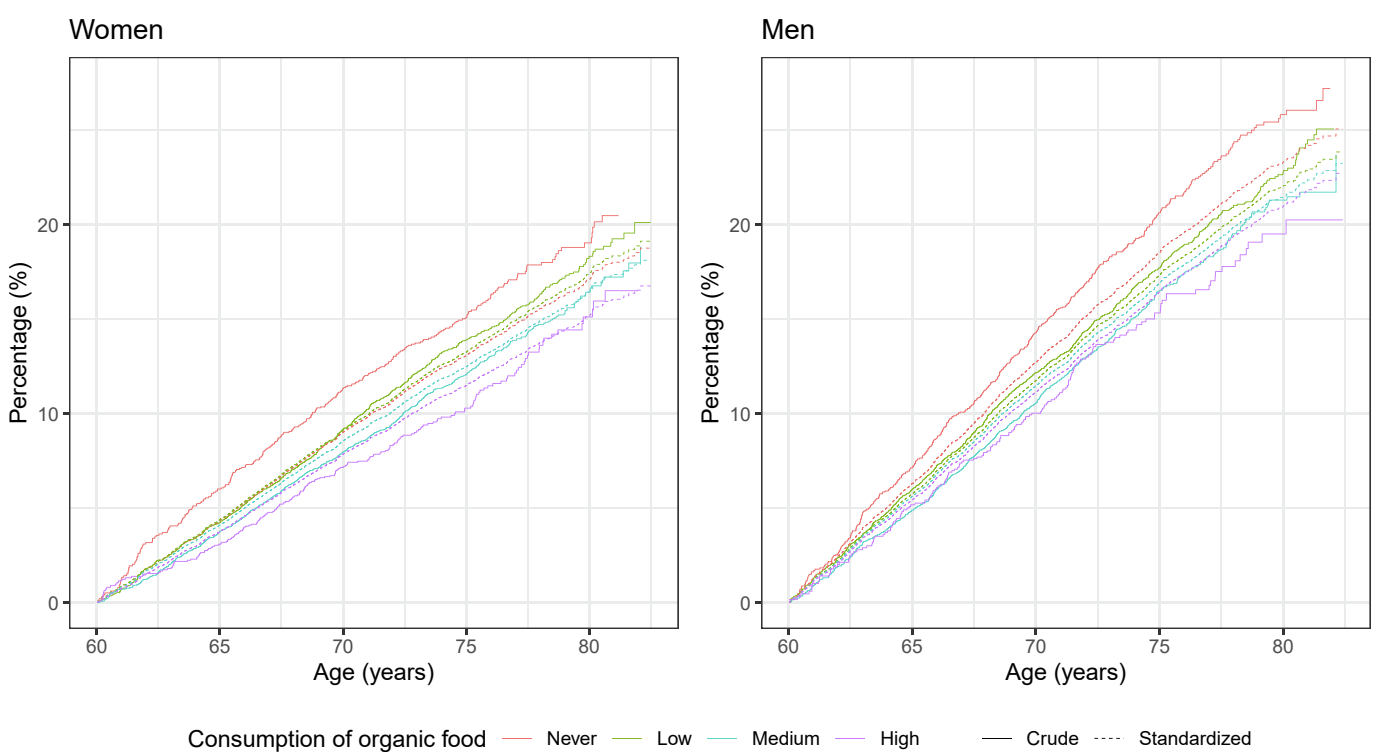

Fig. 2. Crude and average standardized absolute risk of type 2 diabetes mellitus for women and men according to organic food score consumption in four categories: never=6, low intake=7–12, medium intake=13–18 and high intake=19–24. Standardized to the distribution of smoking, physical activity, educational level, BMI, and alcohol consumption by organic food score consumption group.

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also found that purchasing organic eggs most of the time/always was associated with a lower occurrence of diabetes compared to never purchase of organic eggs.

Pesticides are of less relevance when looking at organic versus conventional produce of animal foods, but some studies have found differences in fatty acid composition between organic versus conventionally produced animal products, with especially a higher content of omega-3 fatty acids [7]. Omega-3 fatty acids have shown to influence beta-cell function and thereby improve insulin secretion [34] however, inconsistent results have been observed in epidemiological studies evaluating associations between omega-3 fatty acids and incidence of type 2 diabetes mellitus [35,36]. No clear differences in the nutritional composition of organic eggs, inside or outside free-range eggs, or battery eggs were observed in the Danish national food database [37]. Similarly, inconsistent results exists regarding associations between general consumption of eggs and incidence of type 2 diabetes mellitus, with some studies observing a higher incidence and some with zero findings [38]. Moreover, the median intake of eggs in the current study was 18 g/day corresponding to 1/3 egg per day. It is questionable whether such a small amount and the associated potential difference in fatty acid composition can influence health risks. Adding up all these factors indicates that the finding of a lower incidence of type 2 diabetes mellitus with higher consumption of organic eggs may be a marker of something else. In addition, we did not observe strong associations with organic food consumption of the plant based food groups and risk of diabetes, where pesticides are of relevance. It is also plausible that a high overall organic food consumption frequency is a marker of a healthy and/or wealthy lifestyle and we can therefore not exclude the risk of residual confounding from unidentified factors e.g. household income and occupation or by insufficient adjustment of e.g. smoking. Unfortunately, we do not have information of these covariates.

It has previously been demonstrated that the current cohort participants have higher socioeconomic status compared to non-participants and they were living in one of the two largest cities of Denmark (Copenhagen and Aarhus). However, we do not expect the associations between intake of organic foods and diabetes to be different among participants and non-participants.

Though impact of bias cannot be ruled out, we observed that overall organic food consumption was associated with a suggested lower incidence of type 2 diabetes mellitus in a cohort of Danish adults.

Author contributions

AT and KO were responsible for collection of data.JLMA, KF, CK, JH, OR-N, KO, AT and AO contributed to the design of this study. JLMA performed the statistical data analyses under supervision of KF. JLMA and KF had access to the final dataset for the study, and takes responsibility for the integrity of the dataset and the accuracy of the data analysis. JLMA wrote the first draft of the manuscript with contributions from AO. All authors contributed to interpretation of the results and critical revision of the initial manuscript, and approved the final version.

Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.diabres.2023.110972.

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